

Pankaj Patel

Full-Stack & Systems Engineer | AI, Real-Time & Distributed Backends

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SUMMARY

Full-stack engineer (SDE-3) working across the stack — React and Next.js on the front, Node.js and Python on the back. At VideoSDK I helped grow the platform from **thousands to millions of live participants**, **built the SIP system from scratch** for audio and video calls, and set up the infrastructure that runs **AI voice agents** in production. I also write **Go, Rust and C/C++**, and enjoy the hard parts: finding memory leaks, fixing race conditions, and keeping large systems stable under load on AWS, GCP, Docker and Kubernetes.

TECHNICAL SKILLS

Languages | TypeScript, JavaScript, Python, Go, Rust, C, C++, Java, SQL

AI & LLM | LLM API integration, AI voice & chat agents, agent orchestration & tool calling, streaming inference, prompt engineering, RAG & vector search, evals & observability

Frontend | React, Next.js, Redux, Zustand, Jotai, Tailwind CSS, Material UI, ShadCN, HTML5, CSS3, real-time dashboards & data visualization

Backend | Node.js, Express, Python (FastAPI/asyncio), Go (goroutines, channels, context), REST, GraphQL, gRPC/Protobuf, Web-Socket, HTTP/2, NATS, MQTT, event-driven microservices

Systems & Media | WebRTC, SIP/VoIP, RTP, MediaSoup, Kurento, SFU/media relay, Twilio SIP trunking, DTMF, BWE, TCP/UDP, Linux kernel modules, GStreamer, Rust/Tokio, concurrency & memory profiling

Data & Cloud | MongoDB (aggregation pipelines), Redis, PostgreSQL, BigQuery, Oracle, AWS (EC2, S3, Lambda), GCP, Docker, Kubernetes, Helm, GitLab CI/CD, Linux

PROFESSIONAL EXPERIENCE

VideoSDK | SDE-3, Full-Stack Developer

Dec 2024 – Present

Surat, India | *Real-time communication platform*

- **Helped scale the platform from thousands to millions of live participants**, rebuilding the signaling and media layers so they stay up when traffic spikes.
- **Built the SIP system from scratch** — signaling, media negotiation and call handling. First at the company to carry **both audio and video over SIP**, opening the door to phone calls and AI voice agents. Also integrated **Twilio SIP trunking** (REFER, re-INVITE, call transfer).
- **Set up the deployment system for AI agents** in Python, so voice and chat agents can join live rooms and phone calls and scale up on their own.
- **Led the removal of Archon**, our old media/signaling service — moved everything it did onto the new setup with no downtime, cutting infrastructure cost and making the system easier to run.
- Built a multi-region **signaling manager on NATS** that finds and cleans up dead sessions on its own, so every node sees the same state.
- Tracked down and fixed **memory leaks, race conditions and CPU bottlenecks** in the signaling and media services, making them faster and far more reliable under load.
- Built the backend in **Node.js with NATS, Redis and MongoDB** for signaling, participants and sessions, plus low-latency media pipelines on **WebRTC, MediaSoup, Kurento and RTP**.
- Wrote custom **Kurento modules** in C++ (DTMF detection, RTP stream splitting) for media features the stock version didn't support.
- Shipped **PK Battles** (forwards media between two rooms so people in different sessions can compete live) and **room switching**, which moves people between sessions without reconnecting.
- Built health checks that **scale workers automatically**, and **live React dashboards** for BWE packets and subscription behaviour that made network issues far quicker to debug.

CodeLink Infotech | Full-Stack Web Developer

Nov 2023 – Dec 2024

Surat, India

- **Led a team of 5 developers** on MERN-stack projects — split up the work, reviewed code, met deadlines and mentored the juniors.
- Built an **HRMS** using Strapi CMS, MongoDB aggregation pipelines and role-based dashboards, replacing a lot of manual HR work.
- Built a **Hotel Management System** in Next.js with custom ShadCN components and built-in WebRTC chat and video.
- Set up coding standards and a review process, which cut down on bugs and kept the codebase consistent.

SELECTED PROJECTS

- **Stock Market Analytics Platform (TradingView-style)** | React, Material UI, WebSocket — live charts and market data updating in real time.
- **Linux Kernel Device Driver** | C, ioctl — character driver with a circular queue passing data between kernel and user space.
- **Custom Kurento Media Modules** | C++, GStreamer, RTP — DTMF detection and RTP stream splitting Kurento didn't support out of the box.

EDUCATION

B.Tech, Computer Engineering | Bhagwan Mahavir University

Surat, India

GPA: 8.8 / 10.0